

ECON 1550 International Finance

Price Levels and the Exchange Rate in the Long Run

Law of One Price (LOOP)

- The LOOP holds if for every good i with dollar price P_{US}^i and euro price P_E^i , we have

$$P_{US}^i = E_{\$/\epsilon} \times P_E^i$$

- This is a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}^i}{P_E^i}$$

Purchasing Power Parity (PPP)

- PPP holds if

$$P_{US} = E_{\$/\epsilon} \times P_E$$

where P_{US} is the US price level and P_E is the euro area price level

- This is also a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}}{P_E}$$

Relative PPP

- Relative PPP holds if

$$\frac{E_{\$/\epsilon,t} - E_{\$/\epsilon,t-1}}{E_{\$/\epsilon,t-1}} = \pi_{US,t} - \pi_{E,t}$$

where π_t is inflation $\pi_t = P_t/P_{t-1} - 1$

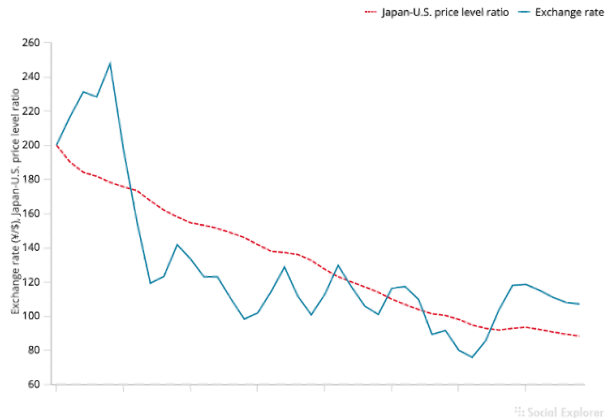
- This is also a theory of exchange rate determination

$$E_{\$/\epsilon,t} = (\pi_{US,t} - \pi_{E,t} + 1)E_{\$/\epsilon,t-1}$$

Relative PPP
does not hold
very well in
the real world

Figure 16-2

The Yen/Dollar Exchange Rate and Relative Japan-U.S. Price Levels, 1980–2019



The graph shows that relative PPP does not track the yen/dollar exchange rate during 1980–2015.

Source: IMF, *International Financial Statistics*. Exchange rates and price levels are end-of-year data.

Problems With PPP

- Price levels of different countries report different baskets of goods
 - E.g. GDP deflator is the basket of goods in GDP for each country
- Deviations from perfectly competitive frictionless markets
 - Transport costs and trade barriers
 - Monopoly power

Expected Relative PPP

- Expected relative PPP (or relative PPP in expectations)

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e$$

where π^e is *expected* inflation $\pi^e = P^e/P - 1$

- This is also a theory of exchange rate determination

$$E_{\$/\epsilon} = \frac{E_{\$/\epsilon}^e}{\pi_{US}^e - \pi_E^e + 1}$$

Relationships

- LOOP $\Rightarrow$ PPP $\Rightarrow$ Relative PPP $\Rightarrow$ Expected relative PPP
- Expected relative PPP + UIP $\Rightarrow$ Fisher effect

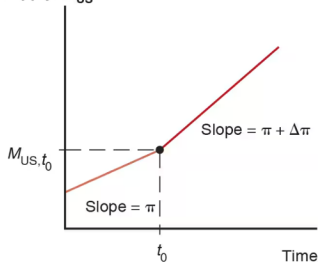
$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e \quad \text{and} \quad R_{\$} = R_{\epsilon} + \frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}}$$

imply

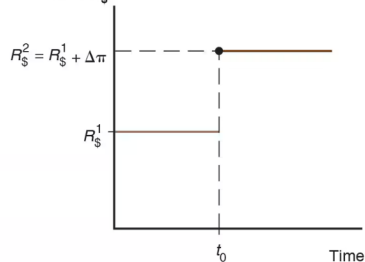
$$R_{\$} - R_{\epsilon} = \pi_{US}^e - \pi_E^e$$

Changes in growth rate of money supply

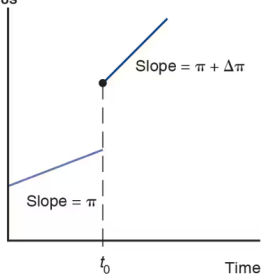
(a) U.S. money
supply, M_{US}



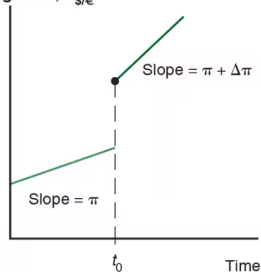
(b) Dollar interest
rate, $R_{\$}$



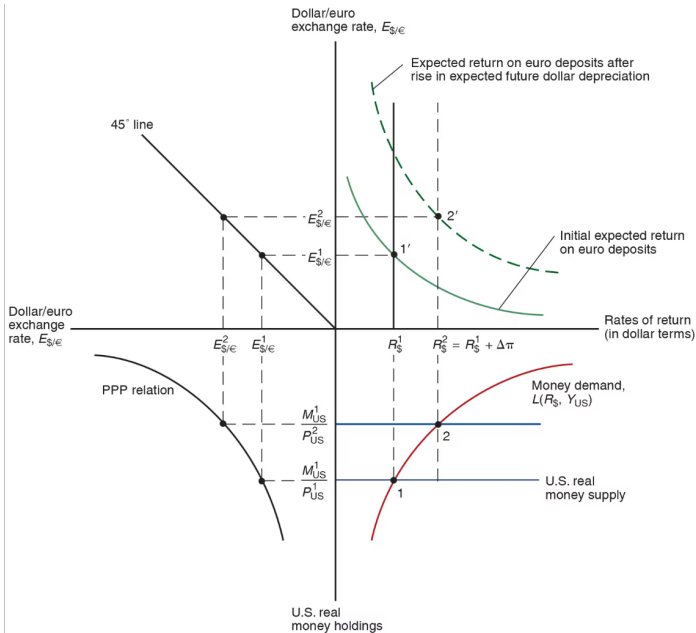
(c) U.S. price
level, P_{US}



(d) Dollar/euro
exchange rate, $E_{\$/\epsilon}$



Changes in growth rate of money supply



Common variables across all models

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
R^*	Foreign interest rate	P	Price level	$M^s/P = L(R, Y)$	Behavioral + eq. condition
M^s, M^{s*}	Money supply	P^*	Foreign price level	$M^{s*}/P^* = L(R^*, Y^*)$	Behavioral + eq. condition

Model 1: PPP + Money Market

Exogenous variables

Variable	Description
R	Domestic interest rate
Y, Y^*	Real incomes

Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$E = P/P^*$	Behavioral

Model 2: Expected Relative PPP + Money Market + UIP

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
Y, Y^*	Real incomes	E	Exchange rate	$R = R^* + E^e/E - 1$	UIP / eq. cond.
g_M, g_M^*	Money supply growth rates	R	Interest rate	$M^s/P = L(R, Y)$	Behavioral + eq. condition
		E^e	Expected exch. rate	$(E^e - E)/E = \pi^e - \pi^{e*}$	Exp. rel. PPP
		r^e, r^{e*}	Expected real int. rates	$r^e = R - \pi^e$	Definition
		π^e, π^{e*}	Expected inflation	$\pi = g_M - g_L$	Behavioral

Model 3: Real Exchange Rate + Output Supply/Demand

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
g_M, g_{M^*}	Money supply growth rates	E	Exchange rate	$R = R^* + E^e/E - 1$	UIP / eq. cond.
		R	Interest rate	$M^s/P = L(R, Y)$	Behavioral + eq. condition
		Y/Y^*	Relative output	$R^D(q) = R^S$	Eq. cond.
		q	Real exchange rate	$q = EP^*/P$	Definition
		r^e, r^{e*}	Expected real int. rates	$r^e = R - \pi^e$	Definition
		π^e, π^{e*}	Expected inflation	$\pi = g_M - g_L$	Behavioral
		E^e	Expected exch. rate	$(E^e - E)/E = \pi^e - \pi^{e*}$	Exp. rel. PPP